ubticareers

TECHNICAL OVERVIEW | Version 1.0

ubticareers connects a recruitment portal with Dynamics 365 Finance & Operations. The current scope covers job listings, candidate applications, private resume uploads, AI-assisted matching and HR review.

Technology stack

Layer	Technology	Purpose
Frontend	React 19, TypeScript, Vite and Tailwind CSS	Public careers portal and HR workspace.
Backend	Python 3.12, FastAPI and Uvicorn	APIs, authentication and integrations.
ERP integration	D365 Finance & Operations custom services; X++	Job requirements, application submission and evaluation result updates.
Metadata database	Supabase PostgreSQL, India region	Application tracking, ownership and evaluation processing state.
Resume storage	Azure Blob Storage	Private storage of uploaded resumes.
AI	Azure AI Foundry; GPT-5.4-mini deployment	Resume-to-job evidence evaluation.
Background jobs	Redis and ARQ	Asynchronous evaluation processing.
Web and packaging	Nginx, Docker and Supervisor	Frontend serving, API routing and container process management.
Automation	GitHub Actions; hosted Ubuntu runners	Checks, Docker image publishing and deployment.

Application flow

1. Candidates browse job requirements retrieved from D365 and sign in through LinkedIn OpenID Connect.
2. Candidates confirm their details, upload a PDF/DOCX resume and provide recruitment consent.
3. The backend stores the resume in private Blob Storage, submits the recruitment record to D365 and records processing metadata in PostgreSQL.
4. A background worker compares the resume against the job requirement and prepares scores, supporting evidence and gaps.
5. Authorized HR users review candidates and results through /hr. Hiring decisions remain with HR.

Architecture

Browser > Azure App Service / Nginx > FastAPI > D365, PostgreSQL and Blob Storage. Redis / ARQ process AI evaluation in the background through Azure AI Foundry.

Azure hosting

A single Docker image contains the built frontend, FastAPI backend, Nginx, ARQ worker and Redis. Supervisor manages these processes. App Service terminates public HTTPS and forwards requests to container port 8080. This is a single-instance deployment approach.

Service	Resource	Purpose
Resource group	ubti-hrprocess-rg	Groups the project Azure resources.
Linux App Service	ubtihrprocess	Hosts the application container.
App Service Plan	ubti-hrprocess	Provides hosting compute.
Container Registry	hrprocess; image repository ubticareers	Stores versioned deployment images.
Blob Storage	Storage account hrprocess; private container hrprocess	Stores candidate resumes.
AI Foundry deployment	gpt-5.4-mini	Provides model inference; end-to-end integration requires confirmation.

CI/CD approach

Source repository: https://github.com/d365-hrprocess/hrprocess

1. A push to main triggers backend/frontend checks and browser smoke tests on a GitHub-hosted Ubuntu runner.
2. After successful checks, the deployment workflow authenticates to Azure using GitHub OIDC federation.
3. Docker builds the verified commit and pushes the image to ACR, tagged with its Git commit.
4. The workflow updates the App Service main container using the image digest and restarts it.
5. The pipeline checks /healthz for application readiness and the expected deployed revision.

Access and configuration

Identity	Name / configuration	Required access
GitHub deployment identity	ubticareers-github-actions	Reader + AcrPush on ACR; Website Contributor on App Service.
App Service managed identity	Identity of ubtihrprocess	AcrPull on ACR to download deployment images.
Application runtime	App Service environment settings	D365, LinkedIn, PostgreSQL, storage and AI credentials/configuration.

The frontend receives no integration secrets. Runtime credentials are held in App Service settings; PostgreSQL uses TLS certificate verification. Blob access is private. GitHub federation avoids a stored Azure client secret. The HR route requires server-enforced authorization.

Verification status

Hosting was reported operational. CI/CD is configured; the latest supplied run passed Azure login and stopped at ACR discovery. A successful automated deployment and end-to-end model evaluation have not yet been confirmed.